

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

E-COMMERCE ORDER MANAGEMENT SYSTEM

Task

Requirement Analysis and Customer Database Module

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the requirements of the E-Commerce Order Management System. The system is designed to manage customers, products, orders, payments, and order status efficiently. This document describes the functional and non-functional requirements of the system and provides a clear understanding of the system to developers and users.

1.2 Scope

The E-Commerce Order Management System is designed to support online shopping and order management. Customers can register, log in, view products, add products to the cart, place orders, and track their orders. The system also maintains customer information and order details in a database. Administrators can manage products, customers, orders, and order status.

1.3 Objectives

The main objectives of the system are:

- To provide a simple and user-friendly online shopping system.
- To manage customer information securely.
- To maintain product and order details efficiently.
- To allow customers to place and track orders.
- To help administrators manage customers, products, and orders.
- To reduce manual work and improve order management accuracy.

2. BUSINESS REQUIREMENTS

The E-Commerce Order Management System is developed to simplify and manage the complete online ordering process. The system should provide a convenient platform for customers and an efficient management system for administrators.

2.1 Business Requirements

BR-1: Customer Registration

The system should allow new customers to create an account by providing their basic details such as name, email, phone number, and address.

BR-2: Customer Login

The system should allow registered customers to securely log in to their accounts.

BR-3: Product Management

The system should allow administrators to add, update, view, and remove product information.

BR-4: Product Browsing

Customers should be able to view available products, prices, descriptions, and product availability.

BR-5: Shopping Cart

Customers should be able to add products to their cart, update quantities, and remove products before placing an order.

BR-6: Order Management

The system should allow customers to place orders and should maintain complete order information in the database.

BR-7: Payment Management

The system should support recording and managing payment information related to customer orders.

BR-8: Order Tracking

Customers should be able to check the current status of their orders, such as Pending, Confirmed, Shipped, and Delivered.

BR-9: Customer Database Management

The system should securely store and manage customer information, including customer details and order-related information.

BR-10: Administrator Management

Administrators should be able to manage customers, products, orders, payments, and order status through the system.

2.2 Business Goals

- To provide a simple and user-friendly online shopping experience.
- To reduce manual work in order processing.
- To maintain accurate customer and order records.
- To improve order processing efficiency.
- To provide secure management of customer information.
- To help administrators monitor and manage orders effectively.

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe the specific functions that the E-Commerce Order Management System must perform.

FR-1: Customer Registration

The system shall allow new customers to create an account by entering their name, email address, phone number, password, and delivery address. The system shall validate the information before creating the customer account.

FR-2: Customer Login

The system shall allow registered customers to log in using their email address and password. The system shall verify the login details and provide access to the customer's account.

FR-3: Product Management

The system shall allow administrators to add, update, view, and delete product information. Product details shall include product name, description, price, category, and availability.

FR-4: Shopping Cart Management

The system shall allow customers to add products to the shopping cart, change product quantities, view the total amount, and remove products from the cart.

FR-5: Order Placement

The system shall allow customers to place an order after selecting products and providing the required delivery details. The system shall generate a unique order ID for each order.

FR-6: Payment Management

The system shall record payment details associated with customer orders. The system shall maintain the payment status, such as Pending, Successful, or Failed.

FR-7: Order Tracking

The system shall allow customers to view the current status of their orders. The order status may include Pending, Confirmed, Shipped, and Delivered.

FR-8: Customer and Order Management

The system shall allow administrators to view and manage customer information and order details. Administrators shall be able to update order status and maintain accurate customer and order records.

4. CUSTOMER DATABASE MODULE

The Customer Database Module is responsible for storing and managing customer information in the E-Commerce Order Management System. It helps the system maintain accurate customer records and supports customer account and order-related activities.

4.1 Database Operations

The Customer Database Module shall support the following operations:

- Create: Add a new customer record to the database.
- Read: Retrieve customer information when required.
- Update: Modify customer information when necessary.
- Delete: Remove a customer record when permitted.
- Search: Find customer records using Customer ID, email, or phone number.

4.3 Data Security

Customer information shall be protected from unauthorized access. Passwords should be stored securely using encryption or hashing techniques. Only authorized users should be allowed to access or modify customer records.

4.2 Customer Table

Field Name	Data Type	Description	Constrains
Customer_ID	INT	Unique identification number of the customer	Primary key
First_Name	VARCHAR(50)	Customers first name	NOT NULL
Last_Name	VARCHAR(50)	Customers last name	NOT NULL
Email	VARCHAR(100)	Customers email address	UNIQUE,NOT NULL
Phone_Number	VARCHAR(15)	Customers contact number	NOT NULL
Password	VARCHAR(255)	Encrypted customer password	NOT NULL
Address	VARCHAR(255)	Customers delivery address	NOT NULL
City	VARCHAR(50)	Customers city	NOT NULL
State	VARCHAR(50)	Customers state	NOT NULL
Pincode	VARCHAR(10)	Customers postal code	NOT NULL
Created_Date	DATE	Date when the customer account was created	NOT NULL
Account_Status	VARCHAR(20)	Current status of the customer account	DEFAULT 'Active'

5. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality, performance, security, and usability expectations of the E-Commerce Order Management System.

NFR-1: Performance

The system should respond quickly to customer requests and process orders efficiently without unnecessary delays.

NFR-2: Security

The system shall protect customer information and prevent unauthorized access. User passwords and sensitive information should be stored securely.

NFR-3: Reliability

The system should operate reliably and maintain accurate customer, product, and order information.

NFR-4: Availability

The system should be available to customers and administrators whenever the service is required, except during planned maintenance.

NFR-5: Usability

The system should have a simple, clear, and user-friendly interface so that customers can easily browse products and place orders.

NFR-6: Scalability

The system should be capable of handling an increasing number of customers, products, and orders as the business grows.

NFR-7: Maintainability

The system should be easy to maintain, update, and modify when new features or changes are required.

NFR-8: Data Integrity

The system shall maintain accurate and consistent customer, product, payment, and order data.

NFR-9: Backup and Recovery

The system should provide suitable data backup and recovery mechanisms to prevent permanent data loss.

NFR-10: Compatibility

The system should work properly on commonly used web browsers and different devices such as laptops, tablets, and mobile phones.

6. SYSTEM USERS AND MODULES

6.1 System Users

The E-Commerce Order Management System mainly consists of two types of users:

1. Customer

Customers can:

- Register and create an account.
- Log in securely.
- Browse available products.
- Add products to the shopping cart.
- Place orders.
- View payment status.
- Track order status.
- Manage their personal information.

2. Administrator

Administrators can:

- Manage customer accounts.
- Add, update, and remove products.
- View and manage customer orders.
- Update order status.

- Manage payment information.
- Monitor the overall order management process.

6.2 System Modules

The major modules of the system are:

1. Customer Management Module

Manages customer registration, login, profile information, and customer records.

2. Product Management Module

Manages product details, categories, prices, and product availability.

3. Shopping Cart Module

Allows customers to add, remove, and update products before placing an order.

4. Order Management Module

Handles order placement, order details, order IDs, and order status.

5. Payment Management Module

Maintains payment information and payment status related to customer orders.

6. Order Tracking Module

Allows customers to view the current status of their orders.

7. Customer Database Module

Stores and manages customer information securely in the database.

7. SYSTEM REQUIREMENTS

7.1 Hardware Requirements

The system can be developed and used with the following basic hardware:

- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 4 GB
- Storage: Minimum 10 GB free space
- Keyboard and Mouse
- Internet Connection

7.2 Software Requirements

The system may require the following software:

- Operating System: Windows, Linux, or macOS
- Web Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox
- Database Management System: MySQL or equivalent
- Programming Environment: Any suitable web development environment
- Code Editor: Visual Studio Code or equivalent
- Version Control: Git and GitHub

7.3 Database Requirements

The database should store and manage:

- Customer information
- Product information
- Shopping cart details
- Order information
- Payment information
- Order status

The database should maintain data accuracy, consistency, and security.

8. CONCLUSION

The E-Commerce Order Management System provides an organized way to manage customers, products, orders, payments, and order tracking. The Requirement Specification defines the major business, functional, database, and non-functional requirements of the system. The Customer Database Module provides a structured approach for storing and managing customer information securely. This document can serve as a reference for the development and implementation of the system.